

ACCURA 3300E

High Accuracy Digital Power Meter

Installed at multiple locations within a facility
actually makes possible power measurement



Communication User Guide[English]

Revision 1.20 2019/1/23

Notice

Symbol

Caution



Indicates the presence of dangerous voltage which can cause severe injury or death to persons if proper precautions are not followed.

Warning



Alerts the user to the presence of hazards which can cause somewhat injury to persons, damage to property or damage to the device itself, if proper precautions are not followed.

Note



Indicates the user's attention to installation, operating and maintenance instructions.



Indicates alternative voltage or current.



Indicates direct voltage or current.

Installation Consideration

Installation and operation of Accura 3300E should be performed only by qualified, competent personnel that have appropriate training and experience with high voltage and current devices.



Caution

Failure to observe the following instructions may result in severe injury or death.

- During normal operation of Accura 3300E, hazardous voltages are present on its terminal strips of voltages input and power.
- Standard safety precautions are as followed while performing any installation or service work [removing PT fuses, shorting CT secondaries, etc.].
- Do not access to terminal strips of Accura 3300E after installation.

**Warning**

Observe the following instructions, or permanent damage to the meter may occur

- Do not apply Accura 3300E to voltages and currents that exceed Input ratings of PT and CT.
- To use device in the other way than specified by manufacturer can cause severe damage.
- Connect ground terminal to the earth ground to protect device from noise and surge.
- Terminal screw torque is as followed.
 - Barrier-type voltage terminal: 1.35Nm[1.00 ft-lbf] max.
 - Barrier-type digital inputs/digital output terminal: 0.90Nm[0.66 ft.lbf] max.

About Manual

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Standard Compliance



Process Control Equipment
E324900



MSIP-REM-RTE-ACCURA 3300E

Revision History

The following versions of Accura 3300E Communication User Guide have been released.

Revision	Date	Description
Revision 1.00	2016. 02. 19	Initial draft
Revision 1.01	2016. 03. 07	Error correction for Register number of Harmonics data
Revision 1.02	2016. 05. 31	Elimination setting of measureable minimum Electrical Power. Elimination due to unprovision of MIN/MAX Power factor.
Revision 1.10	2018. 2. 9	Format changed and bookmark added
Revision 1.20	2019. 1. 23	zero voltage/current --> residual voltage/current word changed

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Chapter 1 Modbus Protocol of Accura 3300E

Modbus Protocol Overview

In this section, we will introduce the fundamentals of the Modbus RTU protocol that are used in Accura 3300E. For more details on the definition of Modbus protocol and Modbus RTU protocol, see www.modbus.org.

Modbus Protocol

The Modbus protocol is the application protocol that defines the rules for organizing and interpreting data that are independent of the data transmission means. The Modbus protocol establishes the format for master's request by placing into it a slave device (or broadcast) address, a function code defining the requested action, any data to be sent, and an error-checking field. The slave's response message is also constructed using Modbus protocol. It contains fields confirming the action taken, any data to be returned, and an error-checking field. If an error occurred when receiving a message, or if the slave is unable to perform the requested action, the slave will construct an error message and send it as its response.

Modbus RTU Protocol

The Modbus RTU protocol is a version of Modbus protocol that is designed to work under a serial communication environment such as RS-485, RS-232, etc. This protocol uses the pre-set device address to identify each device and CRC to check the errors. It does not allow multiple accesses through serial communications.

Modbus Packet Structure

Modbus RTU Packet Structure

The request and response packets of Modbus RTU are composed of the following fields.

Device Address	Function Code	Data	CRC
1 bytes	1 byte	n bytes	2 bytes

The meaning of each field is as follows.

Fields	Description
Device Address	Device address field is used to identify each slave device and can have any value within the range of 1 to 247.
Function Code	When a request is sent from master to slave device, this function code field tells the slave device what kind of action to perform. In the case of normal response, the function code of the corresponding request is used without alteration. In the case of error response, 80h is added to the request function code.
Data	The data fields differ depending on the function code.
CRC	This is an error-checking field that is based on a Cyclical Redundancy Check (CRC) method. The CRC field checks the contents of the entire message. It is generated using CRC-16 (Modbus) algorithm. For more details, see Appendix B.

Accura 3300E Modbus Supports

Function code

Accura 3300E supports the following function codes.

Function code Decimal [Hexadecimal]	Name	Description
3 [03h]	Read Holding Registers ¹	Reads the contents of holding registers 1 - 65536 in the slave device. The request message specifies the starting register and the quantity of registers to be read. Registers are addressed starting at zero: registers 1 - 16 are addressed as 0 - 15.
6 [06h]	Write Single Register	Writes a value to a single holding register, one of 1 - 65536 in the slave device. The request message specifies the holding register reference to be written. Registers are addressed starting at zero: register 1 is addressed as 0. The requested preset value is specified in the request data field.
16 [10h]	Write Multiple Registers	Writes values to multiple consecutive holding registers 1 - 65536 in the slave device. The request message specifies the starting holding register to be written. Registers are addressed starting at zero: register 1 is addressed as 0. The requested preset values are specified in the request data field.

1. Holding register is 16-bit (2-byte) word.

Accura 3300E Function Code Packet Structure

The detailed packet structure of each function code provided by Accura 3300E is as follows.

Function 3 [03h]: Read Holding Registers

This function code can read some parts of the holding registers 1 - 65536 in the slave device.

Each holding register is 2-byte word.

Request

Function Code	Starting Address	Quantity of Registers
1 byte	2 bytes	2 bytes

Response

Function Code	Byte Count	Register Values
1 byte	1 byte	2 * (Quantity of Registers) bytes

Error Response

Error Code	Exception Code
1 byte	1 byte

Detailed structure of Request

Name	Byte Length	Description
Function Code	1	3 [03h]: Read holding registers
Starting Address	2	Starting address to be read. Registers are addressed starting at zero: Register address is obtained by subtracting by 1 from register number of Modbus Map. Holding registers 1 - 65536 are addressed as 0 - 65535.
Quantity of Registers	2	Number of registers to be read. Standard range: 1 - 125

Detailed structure of Response

Name	Byte Length	Description
Function Code	1	3 [03h]: Read holding registers
Byte Count	1	2 * (Quantity of Registers)
Register Values	2 * Quantity of Registers	Contents of holding registers to be read. Contents of holding registers are described in the Modbus Map.

Detailed structure of Error Response

Name	Byte Length	Description
Error Code	1	131 [83h]: Error response of "Read Holding Registers".
Exception Code	1	2: When the holding register number to be read exceeds the limitation of 65536 3: When the requested "Quantity of Registers" is 0 or above 126.

Function 6 [06h]: Write Single Register

This function code can write content to a holding register in the range of 1 - 65536.

Each holding register is 2-byte word.

Request

Function Code	Register Address	Register Value
1 byte	2 byte	2 byte

Response

Function Code	Register Address	Register Value
1 byte	2 byte	2 byte

Detailed structure of Request

Name	Byte Length	Description
Function Code	1	6 [06h]: Write Single Register
Register Address	2	Holding register address to be written. Register is addressed starting at zero: Register address is obtained by subtracting by 1 from register number of Modbus Map. Holding registers 1 - 65536 are addressed as 0 - 65535.
Register Value	2	Content of holding register to be written. Contents of holding registers are described in the Modbus Map.

Detailed structure of Response

Name	Byte Length	Description
Function Code	1	6 [06h]: Write Single Register
Register Address	2	The same value as that of Request packet.
Register Value	2	The same value as that of Request packet.

Function 16 [10h]: Write Multiple Registers

This function code can write contents to some parts of the holding registers 1 - 65536.

Each holding register is 2-byte word.

Request

Function Code	Starting Address	Quantity of Registers	Byte Count	Register Values
1 byte	2 byte	2 byte	1 byte	2 * (Quantity of Registers) bytes

Response

Function Code	Starting Address	Quantity of Registers
1 byte	2 byte	2 byte

Error Response

Error Code	Exception Code
1 byte	1 byte

Detailed structure of Request

Name	Byte Length	Description
Function Code	1	16 [10h]: Write Multiple Registers
Starting Address	2	Starting address to be written. Registers are addressed starting at zero: Register address is obtained by subtracting by 1 from register number of Modbus Map. Holding registers 1 - 65536 are addressed as 0 - 65535.
Quantity of Registers	2	Number of registers to be written. Valid quantity number: 1 - 123
Byte Count	1	2 x Quantity of Registers
Register Values	2 * Quantity of Registers	Contents of holding registers to be written. Contents of holding registers are described in the Modbus Map.

Detailed structure of Response

Name	Byte Length	Description
Function Code	1	16 [10h]: Write Multiple Registers
Starting Address	2	The same value as that of Request packet.
Quantity of Registers	2	The same value as that of Request packet.

Detailed structure of Error Response

Name	Byte Length	Description
Error code	1	144 [90h]: Error response of "Write Multiple Registers"
Exception code	1	2: When the holding register number to be read exceeds the limitation of 65536 3: When the requested "Quantity of Registers" is 0 or above 124.

Chapter 2 Modbus Map of Accura 3300E

Modbus Map Overview

Accura 3300E Modbus Map is composed of five categories: System information, Setup, Control, Measurement data, Event data.

Holding register is addressed starting at zero. Register address is obtained by subtracting by 1 from register number of Modbus Map. Holding register 1 - 65536 are addressed as 0 - 65535.

Register Number	Description
System Information	
10000-10019	Accura 3300E System Information
Setup	
10100	Remote Setup Unlock
10110-10129	User Area Setup
10150-10176	Measurement Setup
10200-10209	User Interface Setup
10250-10255	Serial Communication Setup
10350-10358	Measurement Event Setup
10400-10430	Energy Level Setup
11000-11002	Aggregation Setup
Control	
10300	Remote Control Unlock
10310-10314	Measurement Control
Measurement Data	
11010-11293	Measurement data [Average of aggregation interval]
12000-12197	Measurement Max/Min data [Max/Min of aggregation interval]
13000-13384	Harmonics of voltage and current 0-31th components from 64-sample/cycle per phase
14000-14793	Waveform of voltage and current 64-sample/cycle * 2-cycle * 3-phase
15000-15209	Measurement Max/Min data [After max/min user-reset]
Measurement Event Data	
10380-10388	Measurement Event Data

Data Format

Data Format	Description	Word Length	Endian	Range
UInt16	Unsigned 16-bit	1	NA ¹	0 to 65,535
Int16	Signed 16-bit	1	NA ¹	-32,768 to 32,767
UInt32	Unsigned 32-bit	2	Big-Endian ²	0 to 4,294,967,295
Int32	Signed 32-bit	2	Big-Endian ²	-2,147,483,648 to 2,147,483,647
Float32	Single-precision Float (IEEE 754)	2	Big-Endian ²	-3.4x10 ³⁸ to 3.4x10 ³⁸

1. NA (Not Available): 1-word data, independent of endian.

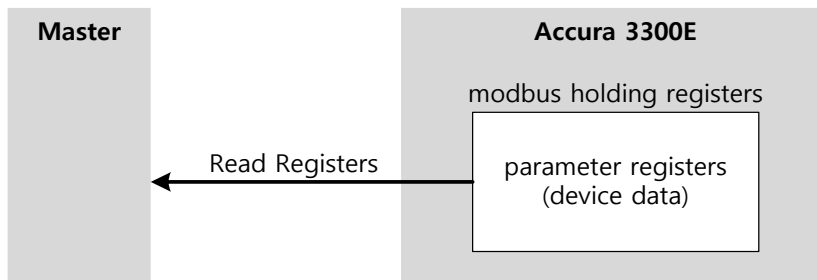
2. 2-word data, two register spaces are used. The upper word is located in the lower address register and the lower word is located in the higher address register.

Data Attribute of Register Access

R: Read Access

Master can obtain device data from the holding registers that have "read attribute" by just reading the holding registers because the holding registers have the device data directly.

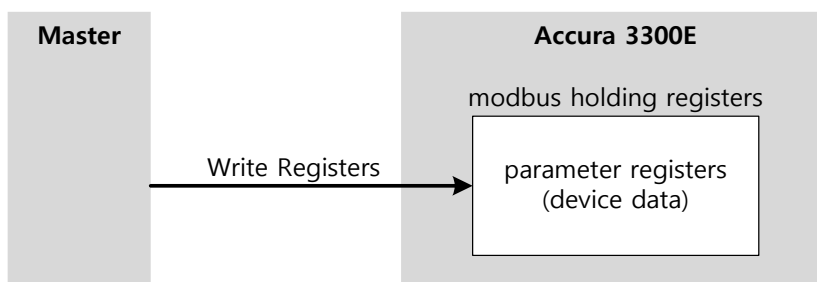
Fig 2.1 Read access of holding register



W: Write Access

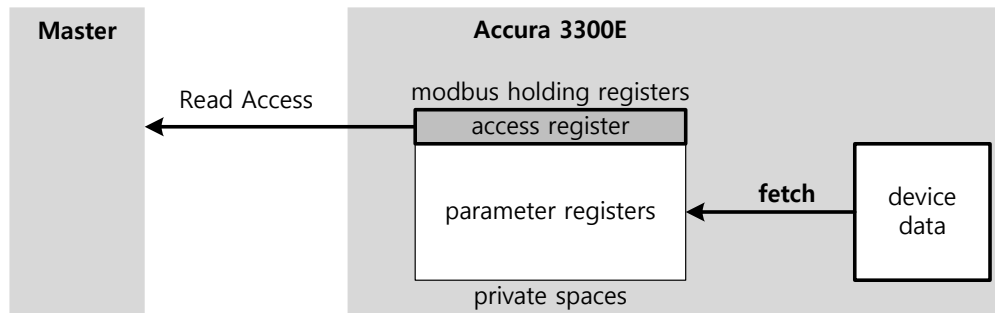
Master can transfer data to the holding registers that have "write attribute" by just writing them to the holding registers that are applied to the device data directly.

Fig 2.2 Write access of holding register

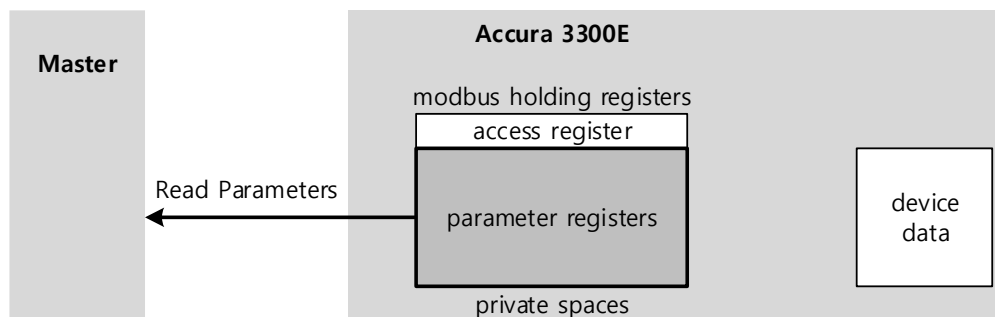


PR: Private Read Access

In order to read a group of data from a device that has a same time-stamp safely for synchronism, two steps are needed. At first, master must fetch them into the parameter registers before reading them. By reading "access register" of the corresponding parameter registers, the device's data is fetched into the parameter registers on the private spaces for the connection.

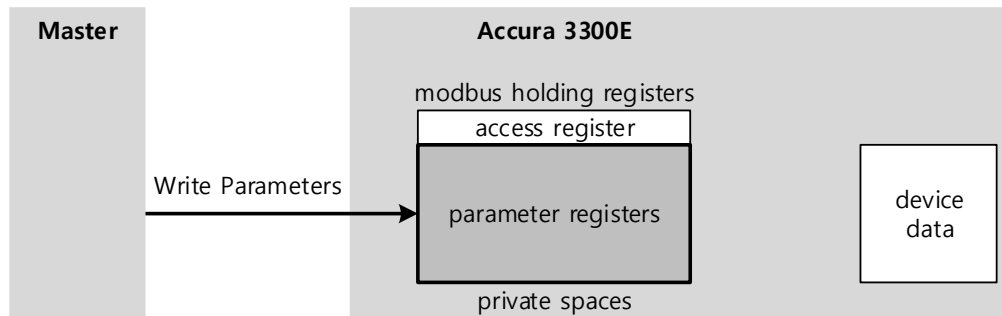
Fig 2.3 Private Read access of holding register: Read Access

After fetching them into the parameter registers (private spaces for the connection), master can read the private spaces safely although the device real data changes during the read interval. The synchronism of the device data obtained is guaranteed because the parameter registers on the private spaces do not change during the read interval.

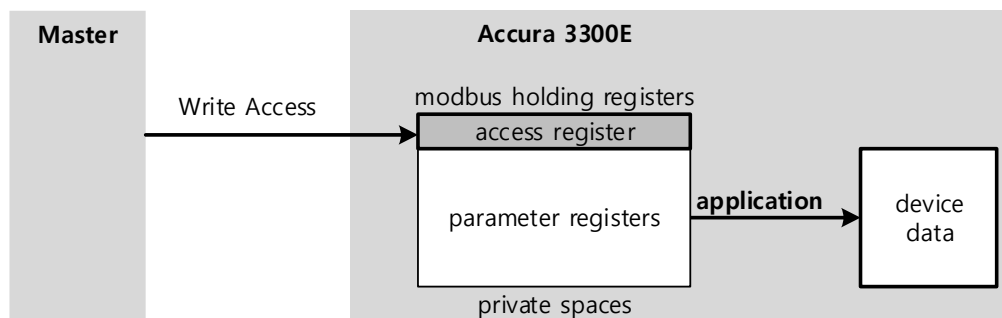
Fig 2.4 Private Read access of holding register: Read Parameter

PW: Private Write Access

In order to write a group of data to be applied at the same time safely, two steps are needed. At first, a group of data must be written into the parameter registers on the private spaces for the connection.

Fig 2.5 Private Write access of holding register: Write Parameter

And then by writing 1 to "access register" of the corresponding parameter registers, the stored data into the parameter registers is applied to the device at the same time for synchronism.

Fig 2.6 Private Write access of holding register: Write Access**RW: Read / Write**

RW attribute means "Read" and "Write" as mentioned above.

PRW: Private Read / Private Write

PRW attribute means "Private Read" and "Private Write" as mentioned above.

System Information Category

Accura 3300E System Information

Register Number	Name	Format	Attribute	Description
10000	Product ID	Uint16	R	Accura 3300E Product ID: 3301
10001	Serial number	Uint32	R	Accura 3300E Serial Number
10003-10012	Vendor name	20*char	R	Manufacturing company. Default: ROOTECH Inc.
10013	Hardware version	Uint16	R	Hardware version
10014	Firmware version	Uint16	R	Firmware version
10015	Map version	Uint16	R	Modbus Map version
10016	Bootloader version	Uint16	R	Bootloader version
10017	Pcb1 ID	Uint16	R	Pcb1 identity number
10018	Pcb2 ID	Uint16	R	Pcb2 identity number
10019	CT type	Uint16	R	CT 5A or CT1A

Setup Category

Setup in remote mode is locked by default. At first, it is necessary to unlock the setup at the Modbus connection in which the change is to be made. Also, another setup unlock is necessary when making new Modbus connection.

Remote Setup Unlock

Register Number	Name	Format	Attribute	Description
10100	Remote setup unlock	UInt16	RW	For setup to be allowed, write these four numbers on this register in the order given below ¹. 2300 → 0 → 700 → 1 ¹ For setup to be locked, write any value on this register. Remote setup lock status can be known by reading this register as shown below. 0: Setup allowed 1: (default) Setup lock

1. If the order is wrong, the entire writing process should start from the beginning again.

User Area Setup

Register Number	Name	Format	Attribute	Description
10110-10129	User area	20* UInt16	RW	Readable/writable space for the user information The values written in this space is preserved in Accura 3300E

Measurement Setup

Register Number	Name	Format	Attribute	Description
10150	Measurement setup access	UInt16	PRW	Access register of Register 10151 - 10176. By reading this register, Fetched from Accura 3300E data to register 10151 - 10176. Bit.[15] of this register is read as 1 on success of fetch. By writing 1 to this register, Applied from register 10151 - 10176 to Accura 3300E.
10151	Wiring mode	UInt16	PRW	Wiring mode. 0: 1P2W, 1-phase 2-wire voltage connection 1: 1P3W, 1-phase 3-wire voltage connection 2: 3P3W, 3-phase 3-wire voltage connection 3: (default) 3P4W, 3-phase 4-wire voltage connection
10152	Phase rotation sequence	UInt16	PRW	Phase rotation sequence setup. 0-sequence unbalance and negative-sequence unbalance are calculated based on the set rotation sequence. 0: (default) Auto-detection 1: Positive sequence 2: Negative sequence
10153	Minimum measured voltage	UInt16	PRW	Minimum measured value of Accura 3300E input voltage. A voltage less than this value is considered to be 0 voltage. Unit [V] Valid range: 1 - 10 [V] (When external PT is used, PT secondary voltage reference) Default: 5 [V]
10154-10155	Primary voltage	UInt32	PRW	Primary line-line voltage of external potential transformer. High voltages over line-line 500V can be connected through the external potential transformer. Unit [V] Valid range: 1 - 999,999 [V] Default: 380 [V]
10156	Secondary voltage	UInt16	PRW	Secondary line-line voltage of external potential transformer. High voltages over line-line 500V can be connected through the external potential transformer. Unit [V] Valid range: 1 - 999 [V] Default: 380 [V]
10157-10158	Primary current	UInt32	PRW	Primary line current of external current transformer. Can be connected through external current transformer when current is over 5A. Unit [0.1A] Valid range: 1 - 999999 (0.1 - 99999.9 [A]) Default: 500 (50 [A])
10159	Secondary current	UInt16	PRW	Secondary line current of external current transformer. Can be connected through external current transformer when current is over 5A. Unit [0.1A] Valid range: 1 - 999 (0.1 - 99.9 [A]) Default: 50 (5 [A])

10160	Power source selection for Demand evaluation	UInt16	PRW	Type of power to be used in calculating the demand. 0: (default) Received power 1: Net power (Received power-Delivered power)
10161	Number of sub-demand	UInt16	PRW	The number of sub-demands for total demand interval. Valid range: 1 - 12 Default: 1
10162	Sub-demand interval time	UInt16	PRW	Interval time of sub-demand. Demand value is updated at every sub-demand interval time. Unit [min] Valid range: 1 - 60 [min] Default: 15 [min] Total demand interval time = (# of sub-demand) * (sub-demand interval time)
10163	Demand Prediction Response index	UInt16	PRW	Demand Prediction Response. The higher the value is, the faster response speed. The smaller the value is, the better attribute to remove peak. Valid range: 0 - 99 Default: 90
10164	Phase power calculation method	UInt16	PRW	Q (Reactive Power) calculation method. 0: (default) Calculation method of fundamental wave. Q is calculated by using the 90° phase-delayed voltage of fundamental wave. 1: Calculation method of harmonic wave. Q is calculated by using apparent electric power S. S is calculated by using RMS voltage and RMS current.
10165	Total power calculation method	UInt16	PRW	Total power calculation method from phase power. 0: (default) Vector sum 1: Arithmetic sum
10166	TDD Reference selection	UInt16	PRW	Select TDD reference value to calculate current TDD. 0: (default) Use TDD nominal current as TDD reference value. Cf) Use Reference current (register 10173) if TDD nominal current is 0. 1: Use measured peak demand as TDD reference value.
10167-10168	TDD nominal current	UInt32	PRW	Set using current TDD as reference value if value of register 10166 is 0. Use Reference current (register 10173) value as current TDD if this register value is 0. Unit [0.1A] Valid range: 1 - 999999 (0.1 - 99999.9 [A]) Default: 0
10169	Minimum measured current	UInt16	PRW	Measurable minimum current level. Measured current of the device under the minimum measurable current is considered as 0. Unit [mA] Valid range: 1 - 100 [mA] (Apply secondary voltage of potential transformer while use external potential transformer) Default: 20 [mA]
10170	Reserved			

10171-10172	Reference voltage	Uint32	PRW	Primary reference voltage. Display bar graph percentage based on this voltage. Unit [V] Valid range: 1 - 999999 [V] (Primary voltage) Default: 380 [V]
10173-10174	Reference current	Uint32	PRW	Primary reference current. Display bar graph percentage based on this current. Unit [0.1A] Valid range: 1 - 999999 (0.1 - 99999.9 [A]) (Primary current) Default: 500 (50 [A])
10175	Power factor display setup	Uint16	PRW	Power factor display setup. Bit.[0]: Power factor indication value when apparent electric power is 0. 0: (default) PF 1.0 indicate 1: PF 0.0 indicate Bit.[8]: Sign of power factor(positive/negative number indication. 1: (default) sign exist, $PF = P/S$. Positive value means power receiving status, Negative value means power transmission status. 0: sign does not exist, $PF = \text{abs}(P)/S$.
10176	Reactive power sign	Uint16	PRW	Reactive power sign display setup. Bit.[8]: Set sign of reactive power indication. 0: (default) sign exist. Positive value means Inductive reactive power, Negative value means Capacitive reactive power. 1: sign does not exist.

User Interface Setup

Register Number	Name	Format	Attribute	Description
10200	User interface setup access	UInt16	RW	Access register of Register 10201 - 10209. By reading this register, Fetched from Accura 3300E data to register 10201 - 10209. Bit.[15] of this register is read as 1 on success of fetch. By writing 1 to this register, Applied from register 10201 - 10209 to Accura 3300E.
10201	LCD Backlight off timeout	UInt16	RW	The time taken to turn off LCD backlight automatically when there is no button input. Unit [sec] Valid range: 10 - 300 [sec] Default: 60 [sec]
10202	Backlight high duty	UInt16	RW	Setting duty ratio for the highest brightness of backlight. Unit [%] Valid range: 80 - 99 [%] Default: 99 [%]
10203	Backlight middle duty	UInt16	RW	Setting duty ratio for middle brightness of backlight. Unit [%] Valid range: 50 - 80 [%] Default: 99 [%]
10204	Backlight low duty	UInt16	RW	Setting duty ratio for the lowest brightness of backlight. Unit [%] Valid range: 0 - 20 [%] Default: 99 [%]
10205	Setup exit timeout	UInt16	RW	The time taken to exit to display mode automatically when there is no button input at setup mode. Unit [sec] Valid range: 60 - 3600 [sec] Default: 600 [sec]
10206	Energy display type	UInt16	RW	Selection of energy type to be displayed in the VIPE page. 0: kWh Received. Energy flows from source to load 1: kWh Delivered. Energy flows from load to source 2: kWh Sum: (kWh Received) + (kWh Delivered) 3: kWh Net: (kWh Received) - (kWh Delivered)
10207	Local setup lock	UInt16	RW	Local setup at Accura 3300E. 0: (default) Allowed 1: Not allowed
10208	Event backlight time	UInt16	RW	Set blinking time for backlight in event of start event. Unit [min] 0: blinking for 5 seconds 1 - 9999: blinking for 1 - 9999 [min] 10000: blinking stop by user clear Default: 0 (blinking for 5 seconds)
10209	Beep sound enable	UInt16	PRW	Selection of buzzer sound operation. 0: Sound off 1: (default) Sound on

Serial Communication Setup

Register Number	Name	Format	Attribute	Description
10250	Serial communication setup access	UInt16	PRW	Access register of Register 10251 - 10255. By reading this register, Fetched from Accura 3300E data to register 10251 - 10255. Bit.[15] of this register is read as 1 on success of fetch. By writing 1 to this register, Applied from register 10251 - 10255 to Accura 3300E.
10251	Device address	UInt16	PRW	Modbus serial address. (Slave) Valid range: 1 - 247 Default: 0
10252	Baud rate	UInt16	PRW	Baud rate. 0: 1200 1: 2400 2: 4800 3: (default) 9600 4: 19200 5: 38400 6: 57600 7: 115200
10253	Parity	UInt16	PRW	Parity bit. 0: (default) None parity 1: Odd parity 2: Even parity
10254	Stop bits	UInt16	PRW	Stop bit. 0: (default) 1-stop bit 1: 2-stop bit
10255	Protocol selection	UInt16	PRW	0: Rootech protocol 1: (default) Modbus RTU

Measurement Event Setup

Register Number	Name	Format	Attribute	Description
10350	Measurement event setup access	UInt16	PRW	Access register of Register 10351 - 10358. By reading this register, Fetched from Accura 3300E data to register 10351 - 10358. Bit.[15] of this register is read as 1 on success of fetch. By writing 1 to this register, Applied from register 10351 - 10358 to Accura 3300E.
10351	Fuse-fail detection	UInt16	PRW	Fuse-fail detection when voltage is 0 in current flowing state. 0: (default) Disable 1: Enable
10352	Blackout detection	UInt16	PRW	Blackout detection when voltage and current are 0. 0: (default) Disable 1: Enable
10353	Over-temperature detection	UInt16	PRW	Over-temperature detection when measured temperature is over than event start temperature. 0: (default) Disable 1: Enable
10354	Start temperature of event	UInt16	PRW	Start temperature of Over-temperature event. Unit [°C] Valid range: 20 - 200 [°C] Default: 50 [°C]
10355	End temperature of event	UInt16	PRW	End temperature of Over-temperature event. Unit [°C] Valid range: 0 to (Start temperature of event-1) [°C] Default: 48 [°C]
10356	Event LED clear	UInt16	PRW	Event LED cleared by user. 0: Not cleared 1: Cleared (auto-cleared to 0)
10357	Event backlight clear	UInt16	PRW	Event backlight cleared by user. 0: Not cleared 1: Cleared (auto-cleared to 0)
10358	Event log clear	UInt16	PRW	Event log cleared by user. 0: Not cleared 1: Cleared (auto-cleared to 0)

Energy Level Setup

Register Number	Name	Format	Attribute	Description
10400	Energy level setup access	UInt16	PRW	Access register of Register 10401 - 10430. By reading this register, Fetched from Accura 3300E data to register 10401 - 10430. Bit.[15] of this register is read as 1 on success of fetch. By writing 1 to this register, Applied from register 10401 - 10430 to Accura 3300E.
10401	Received kWh of phase A	Int32	PRW	Received energy of Phase A, Active power. Unit [kWh]
10403	Received kWh of phase B	Int32	PRW	Received energy of Phase B, Active power. Unit [kWh]
10405	Received kWh of phase C	Int32	PRW	Received energy of Phase C, Active power. Unit [kWh]
10407	Delivered kWh of phase A	Int32	PRW	Delivered energy of Phase A, Active power. Unit [kWh]
10409	Delivered kWh of phase B	Int32	PRW	Delivered energy of Phase B, Active power. Unit [kWh]
10411	Delivered kWh of phase C	Int32	PRW	Delivered energy of Phase C, Active power. Unit [kWh]
10413	Received kVARh of phase A	Int32	PRW	Received energy of Phase A, Reactive (imaginary) power. Unit [kVARh]
10415	Received kVARh of phase B	Int32	PRW	Received energy of Phase B, Reactive (imaginary) power. Unit [kVARh]
10417	Received kVARh of phase C	Int32	PRW	Received energy of Phase C, Reactive (imaginary) power. Unit [kVARh]
10419	Delivered kVARh of phase A	Int32	PRW	Delivered energy of Phase A, Reactive (imaginary) power. Unit [kVARh]
10421	Delivered kVARh of phase B	Int32	PRW	Delivered energy of Phase B, Reactive (imaginary) power. Unit [kVARh]
10423	Delivered kVARh of phase C	Int32	PRW	Delivered energy of Phase C, Reactive (imaginary) power. Unit [kVARh]
10425	kVAh of phase A	Int32	PRW	Energy of Phase A, Apparent power. Unit [kVAh]
10427	kVAh of phase B	Int32	PRW	Energy of Phase B, Apparent power. Unit [kVAh]
10429	kVAh of phase C	Int32	PRW	Energy of Phase C, Apparent power. Unit [kVAh]

Aggregation Setup

Number	Name	Format	Attribute	Description
11000	Aggregation setup access	UInt16	PRW	Access register of Register 11001 - 11002. By reading this register, Fetched from Accura 3300E data to register 11001 - 11002. Bit.[15] of this register is read as 1 on success of fetch. By writing 1 to this register, Applied from register 11001 - 11002 to Accura 3300E.
11001	Aggregation selection	UInt16	PRW	Selection of measurement data aggregation. 0: (default) Aggregation 1 sec, Avg. included 1: Aggregation 5 sec, Avg/Max/Min included 2: Aggregation 1 min, Avg/Max/Min included
11002	Update mode	UInt16	PRW	Update mode of measured data. 0: Non-fetch mode measurement of recent aggregation interval is automatically updated for every time measurement occurs. 1: (default) Fetch mode measurement of recent aggregation interval is automatically updated only if the register number of being measured value is equal to or smaller than the register number recently read.

Control Category

Remote Control Unlock

Register Number	Name	Format	Attribute	Description
10300	Remote control unlock	UInt16	PRW	For control to be allowed, write these four numbers on this register in the order given below ¹. 2300 → 0 → 1600 → 1 ¹ For control to be locked, write any value on this register. Control-lock status can be known by reading this register as shown below: 0: Control allowed 1: (default) Control locked

1. If the order is wrong, the entire writing process should start from the beginning again.

Measurement Control

Register Number	Name	Format	Attribute	Description
10310	Sub-demand synchronization	UInt16	RW	By writing 1 on this register, the start time of sub-demand interval is synchronized with the current time. This register is auto-cleared internally.
10311	Demand reset	UInt16	RW	By writing a 1 to this register, the demand values of all connected three-phase and single-phase modules are initialized (clear). This register is auto-cleared internally. Peak demand is not cleared by "Demand reset".
10312	Max/Min reset	UInt16	RW	By writing 1 on this register, the maximum/minimum values are cleared. This register is auto-cleared internally. Peak demand is cleared by "Max/Min reset".
10313	Energy reset	UInt16	RW	By writing 1 on this register, the energy values are cleared. This register is automatically set to zero.
10314	Demo mode	UInt16	RW	Demonstration mode 0: (default) Not used demonstration mode. Normal operation mode by voltage and current sensing 1: 3-phase balance mode by internal lookup table 2: 3-phase unbalance mode by internal lookup table

Measurement Data Category

Aggregation Data

In Accura 3300E device, basic measurement data is obtained by processing of sensed voltage and current for every 1 second.

Based on this 1 second data array, aggregation values during longer interval than 0.2 second are obtained

The aggregation values are average and max/min with time-stamp. Average and max/min with time-stamp are generated for each aggregation. Average is the mean value of data array during the aggregation interval. Max/Min is the maximum / minimum value of data array of which resolution is 1 second during the aggregation interval.

Aggregation Data Collect

In Accura 3300E device, the supported aggregation intervals are 0, 1 and 2. (1 second, 5 seconds, 1 minute)

At first, set the aggregation interval you want at "Aggregation selection" register (it is necessary to unlock the setup locking)

Average, maximum and minimum values of aggregation are all the same when aggregation 0[1 second] is selected.

Measurement Data [Average of aggregation interval]

This is the average value of each aggregation interval. When gathering register number 11010 - 12197 by modbus multi-packet, there is no additional data fetch of the packet which has increasing register number for data simultaneity. There is data auto-fetch only for packet which has decreasing register number within the range of register number 11010 - 12197.

Register Number	Name	Format	Attribute	Description
11010	Validity of data	Uint16	R	Register 11011 - 15209 validity. 0: Data invalid 1: Data valid
11011	Aggregation count	Uint32	R	Aggregation count. Valid range: 0 - 4,294,967,295
11013	Phase sequence status	Uint16	R	Bit.[1-0]: Phase sequence 0: Auto detection 1: Positive sequence 2: Negative sequence
11014	Voltage Van	Float32	R	Line to neutral voltage between phase A and neutral. Unit [V]
11016	Voltage Vbn	Float32	R	Line to neutral voltage between phase B and neutral. Unit [V]
11018	Voltage Vcn	Float32	R	Line to neutral voltage between phase C and neutral. Unit [V]
11020	Voltage Vavg_In	Float32	R	Average of 3 line to neutral voltages Van, Vbn, Vcn. Unit [V]
11022	Current Ia	Float32	R	Current of phase A. Unit [A]
11024	Current Ib	Float32	R	Current of phase B. Unit [A]
11026	Current Ic	Float32	R	Current of phase C. Unit [A]
11028	Current Iavg	Float32	R	Average current of 3 phase currents. Unit [A]
11030	Voltage Vab	Float32	R	Line to line voltage between phase A and B. Unit [V]
11032	Voltage Vbc	Float32	R	Line to line voltage between phase B and C. Unit [V]
11034	Voltage Vca	Float32	R	Line to line voltage between phase C and A. Unit [V]
11036	Voltage Vavg_II	Float32	R	Average of 3 line to line voltages Vab, Vbc, Vca. Unit [V]
11038	Active Power Pa	Float32	R	Active power of phase A. Unit [kW]
11040	Active Power Pb	Float32	R	Active power of phase B. Unit [kW]
11042	Active Power Pc	Float32	R	Active power of phase C. Unit [kW]
11044	Active Power Ptot	Float32	R	Total active power of all phases. Unit [kW]
11046	Reactive Power Qa	Float32	R	Reactive power of phase A. Unit [kVAR]
11048	Reactive Power Qb	Float32	R	Reactive power of phase B. Unit [kVAR]
11050	Reactive Power Qc	Float32	R	Reactive power of phase C. Unit [kVAR]
11052	Reactive Power Qtot	Float32	R	Total reactive power of all phases. Unit [kVAR]
11054	Apparent Power Sa	Float32	R	Apparent power of phase A. Unit [kVA]
11056	Apparent Power Sb	Float32	R	Apparent power of phase B. Unit [kVA]
11058	Apparent Power Sc	Float32	R	Apparent power of phase C. Unit [kVA]
11060	Apparent Power Stot	Float32	R	Total apparent power of all phases. Unit [kVA]
11062	PF A	Float32	R	Power factor of phase A.

11064	PF B	Float32	R	Power factor of phase B.
11066	PF C	Float32	R	Power factor of phase C.
11068	Total PF	Float32	R	Total power factor of all phases.
11070	Angle of PF A	Uint16	R	Angle status of phase-A power factor. 0: Invalid (Apparent power is equal to zero) 1: Lead angle 2: Lag angle
11071	Angle of PF B	Uint16	R	Angle status of phase-B power factor. See "Angle of PF A". (register 11070)
11072	Angle of PF C	Uint16	R	Angle status of phase-C power factor. See "Angle of PF A". (register 11070)
11073	Angle of Total PF	Uint16	R	Angle status of total power factor. See "Angle of PF A". (register 11070)
11074	Received kWh	Int32	R	Received kWh energy of total active power. Unit [kWh]
11076	Delivered kWh	Int32	R	Delivered kWh energy of total active power. Unit [kWh]
11078	Sum kWh	Int32	R	Sum kWh energy: Received kWh + Delivered kWh. Unit [kWh]
11080	Net kWh	Int32	R	Net kWh energy: Received kWh - Delivered kWh. Unit [kWh]
11082	Received kVARh	Int32	R	Received kVARh energy of total reactive power. Unit [kVARh]
11084	Delivered kVARh	Int32	R	Delivered kVARh energy of total reactive power. Unit [kVARh]
11086	Sum kVARh	Int32	R	Sum kVARh energy: Received kVARh + Delivered kVARh
11088	Net kVARh	Int32	R	Net kVARh energy: Received kVARh - Delivered kVARh
11090	kVAh	Int32	R	kVAh energy of total apparent power. Unit [kVAh]
11092	Received kWh A	Int32	R	Received kWh energy of phase A, active power. Unit [kWh]
11094	Received kWh B	Int32	R	Received kWh energy of phase B, active power. Unit [kWh]
11096	Received kWh C	Int32	R	Received kWh energy of phase C, active power. Unit [kWh]
11098	Delivered kWh A	Int32	R	Delivered kWh energy of phase A, active power. Unit [kWh]
11100	Delivered kWh B	Int32	R	Delivered kWh energy of phase B, active power. Unit [kWh]
11102	Delivered kWh C	Int32	R	Delivered kWh energy of phase C, active power. Unit [kWh]
11104	Received kVARh A	Int32	R	Received kVARh energy of phase A, reactive power. Unit [kVARh]
11106	Received kVARh B	Int32	R	Received kVARh energy of phase B, reactive power. Unit [kVARh]
11108	Received kVARh C	Int32	R	Received kVARh energy of phase C, reactive power. Unit [kVARh]
11110	Delivered kVARh A	Int32	R	Delivered kVARh energy of phase A, reactive power. Unit [kVARh]
11112	Delivered kVARh B	Int32	R	Delivered kVARh energy of phase B, reactive power. Unit [kVARh]
11114	Delivered kVARh C	Int32	R	Delivered kVARh energy of phase C, reactive power. Unit [kVARh]
11116	kVAh A	Int32	R	kVAh energy of phase A, apparent power. Unit [kVAh]
11118	kVAh B	Int32	R	kVAh energy of phase B, apparent power. Unit [kVAh]
11120	kVAh C	Int32	R	kVAh energy of phase C, apparent power. Unit [kVAh]

11122	Demand Pa	Float32	R	Active power demand of phase A. Unit [kW]
11124	Demand Pb	Float32	R	Active power demand of phase B. Unit [kW]
11126	Demand Pc	Float32	R	Active power demand of phase C. Unit [kW]
11128	Demand Ptot	Float32	R	Total active power demand of all phases. Unit [kW]
11130	Prediction Demand Ptot	Float32	R	Total active power prediction demand of all phases. Unit [kW]
11132	Demand Qa	Float32	R	Reactive power demand of phase A. Unit [kVAR]
11134	Demand Qb	Float32	R	Reactive power demand of phase B. Unit [kVAR]
11136	Demand Qc	Float32	R	Reactive power demand of phase C. Unit [kVAR]
11138	Demand kVAR total	Float32	R	Total reactive power demand of all phases. Unit [kVAR]
11140	Prediction Demand Qtot	Float32	R	Total reactive power prediction demand of all phases. Unit [kVAR]
11142	Demand Sa	Float32	R	Apparent power demand of phase A. Unit [kVA]
11144	Demand Sb	Float32	R	Apparent power demand of phase B. Unit [kVA]
11146	Demand Sc	Float32	R	Apparent power demand of phase C. Unit [kVA]
11148	Demand Stot	Float32	R	Total apparent power demand of all phases. Unit [kVA]
11150	Prediction Demand Stot	Float32	R	Total apparent power prediction demand of all phases. Unit [kVA]
11152	Demand Ia	Float32	R	Current demand of phase A. Unit [A]
11154	Demand Ib	Float32	R	Current demand of phase B. Unit [A]
11156	Demand Ic	Float32	R	Current demand of phase C. Unit [A]
11158	Demand Iavg	Float32	R	Average current demand of all phases. Unit [A]
11160	Prediction demand Iavg	Float32	R	Average current prediction demand of all phases. Unit [A]
11162-11201	Reserved			
11202	Voltage Van1	Float32	R	Fundamental component of line to neutral Van. Unit [V]
11204	Voltage Vbn1	Float32	R	Fundamental component of line to neutral Vbn. Unit [V]
11206	Voltage Vcn1	Float32	R	Fundamental component of line to neutral Vcn. Unit [V]
11208	Voltage Vavg1	Float32	R	Average of 3 fundamental components Van1, Vbn1, Vcn1. Unit [V]
11210	Current Ia1	Float32	R	Fundamental component of phase current Ia. Unit [A]
11212	Current Ib1	Float32	R	Fundamental component of phase current Ib. Unit [A]
11214	Current Ic1	Float32	R	Fundamental component of phase current Ic. Unit [A]
11216	Current Iavg1	Float32	R	Average of 3 fundamental components Ia1, Ib1, Ic1. Unit [A]
11218	Voltage THD A	Float32	R	Total harmonic distortion of line to neutral Van. Unit [%]
11220	Voltage THD B	Float32	R	Total harmonic distortion of line to neutral Vbn. Unit [%]
11222	Voltage THD C	Float32	R	Total harmonic distortion of line to neutral Vcn. Unit [%]
11224	Current THD A	Float32	R	Total harmonic distortion of current Ia. Unit [%]
11226	Current THD B	Float32	R	Total harmonic distortion of current Ib. Unit [%]
11228	Current THD C	Float32	R	Total harmonic distortion of current Ic. Unit [%]
11230	Current TDD A	Float32	R	Total demand distortion of current Ia. Unit [%]

11232	Current TDD B	Float32	R	Total demand distortion of current Ib. Unit [%]
11234	Current TDD C	Float32	R	Total demand distortion of current Ic. Unit [%]
11236	Voltage Phasor Vax	Float32	R	X-coordinate of phasor voltage for phase A. Unit [V]
11238	Voltage Phasor Vay	Float32	R	Y-coordinate of phasor voltage for phase A. Unit [V]
11240	Voltage Phasor Vbx	Float32	R	X-coordinate of phasor voltage for phase B. Unit [V]
11242	Voltage Phasor Vby	Float32	R	Y-coordinate of phasor voltage for phase B. Unit [V]
11244	Voltage Phasor Vcx	Float32	R	X-coordinate of phasor voltage for phase C. Unit [V]
11246	Voltage Phasor Vcy	Float32	R	Y-coordinate of phasor voltage for phase C. Unit [V]
11248	Current Phasor Iax	Float32	R	X-coordinate of phasor current for phase A. Unit [A]
11250	Current Phasor Iay	Float32	R	Y-coordinate of phasor current for phase A. Unit [A]
11252	Current Phasor Ibx	Float32	R	X-coordinate of phasor current for phase B. Unit [A]
11254	Current Phasor Iby	Float32	R	Y-coordinate of phasor current for phase B. Unit [A]
11256	Current Phasor Icx	Float32	R	X-coordinate of phasor current for phase C. Unit [A]
11258	Current Phasor Icy	Float32	R	Y-coordinate of phasor current for phase C. Unit [A]
11260	Residual voltage	Float32	R	Residual voltage by sum of 3-phase voltage. Unit [V]
11262	Residual current	Float32	R	Residual current by sum of 3-phase current. Unit [A]
11264	Voltage Unbalance of Vln	Float32	R	Unbalance ratio of line to neutral voltages. Unit [%] Max. deviation of (Van,Vbn,Vcn) from Vavg_In / Vavg_In * 100 [%]
11266	Voltage Unbalance of Vll	Float32	R	Unbalance ratio of line to line voltages. Unit [%] Max. deviation of (Vab,Vbc,Vca) from Vavg_II / Vavg_II * 100 [%]
11268	Voltage U0 unbalance	Float32	R	Voltage zero-sequence unbalance ratio. Unit [%] zero seq. component / positive seq. component *100 [%]
11270	Voltage U2 unbalance	Float32	R	Voltage negative-sequence unbalance ratio. Unit [%] negative seq. component / positive seq. component *100 [%]
11272	Current Unbalance	Float32	R	Unbalance ratio of 3 phase currents. Unit [%] Max. deviation of (Ia,Ib,Ic) from Iavg / Iavg * 100 [%]
11274	Current U0 unbalance	Float32	R	Current zero-sequence unbalance ratio. Unit [%] zero seq. component / positive seq. component *100 [%]
11276	Current U2 unbalance	Float32	R	Current negative-sequence unbalance ratio. Unit [%] negative seq. component / positive seq. component *100 [%]
11278	CFa	Float32	R	Crest factor of phase A current.
11280	CFb	Float32	R	Crest factor of phase B current.
11282	CFc	Float32	R	Crest factor of phase C current.
11284	KFa	Float32	R	K-factor of phase A current.
11286	KFb	Float32	R	K-factor of phase B current.
11288	KFc	Float32	R	K-factor of phase C current.
11290	Frequency	Float32	R	Frequency of input voltage. Unit [Hz]
11292	Temperature	Float32	R	Temperature measured from rear side of Accura 3300E. Unit [°C]

Measurement Max/Min Data [Max/Min of aggregation interval]

Maximum and minimum value among data measured during 1 second of every selected aggregation interval. In case of reading register number 11010 - 12197 by Modbus multi packet, there is no additional data fetch for the packets of increasing register number in order to preserve synchronization of data.

Data fetch is done automatically only for the packets of decreasing register number about register number 11010 - 12197 region.

Register Number	Name	Format	Attribute	Description
12000	Van max	Float32	R	Max of Line to neutral voltage between phase A and neutral. Unit [V]
12002	Vbn max	Float32	R	Max of Line to neutral voltage between phase B and neutral. Unit [V]
12004	Vcn max	Float32	R	Max of Line to neutral voltage between phase C and neutral. Unit [V]
12006	Vavg_In max	Float32	R	Max of Average of 3 line to neutral voltages Van, Vbn, Vcn. Unit [V]
12008	Vab max	Float32	R	Max of Line to line voltage between phase A and B. Unit [V]
12010	Vbc max	Float32	R	Max of Line to line voltage between phase B and C. Unit [V]
12012	Vca max	Float32	R	Max of Line to line voltage between phase C and A. Unit [V]
12014	Vavg_II max	Float32	R	Max of Average of 3 line to line voltages Vab, Vbc, Vca. Unit [V]
12016	THDva max	Float32	R	Max of Total harmonic distortion of line to neutral Van. Unit [%]
12018	THDvb max	Float32	R	Max of Total harmonic distortion of line to neutral Vbn. Unit [%]
12020	THDvc max	Float32	R	Max of Total harmonic distortion of line to neutral Vcn. Unit [%]
12022	Unbal_VIn max	Float32	R	Max of Unbalance ratio of line to neutral voltages. Unit [%]
12024	Unbal_VII max	Float32	R	Max of Unbalance ratio of line to line voltages. Unit [%]
12026	Unbal_U0_V max	Float32	R	Max of voltage zero-sequence unbalance ratio. Unit [%]
12028	Unbal_U2_V max	Float32	R	Max of voltage negative-sequence unbalance ratio. Unit [%]
12030	Temperature max	Float32	R	Max of Ambient temperature of Accura 3300E device. Unit [°C]
12032	Residual voltage max	Float32	R	Max of residual voltage by sum of 3-phase voltages (Van + Vbn + Vcn). Unit [V]
12034	Frequency max	Float32	R	Max value of frequency. Unit [Hz]
12036	Van min	Float32	R	Min of line to neutral voltage between A and neutral. Unit [V]
12038	Vbn min	Float32	R	Min of line to neutral voltage between B and neutral. Unit [V]
12040	Vcn min	Float32	R	Min of line to neutral voltage between C and neutral. Unit [V]
12042	Vavg_In min	Float32	R	Min of average of 3 line to neutral voltages Van, Vbn, Vcn. Unit [V]
12044	Vab min	Float32	R	Min of line to line voltage between A and B. Unit [V]
12046	Vbc min	Float32	R	Min of line to line voltage between B and C. Unit [V]
12048	Vca min	Float32	R	Min of line to line voltage between C and A. Unit [V]
12050	Vavg_II min	Float32	R	Min of average of 3 line to line voltages Vab, Vbc, Vca.

				Unit [V]
12052	Residual voltage min	Float32	R	Min of residual voltage by sum of 3-phase voltages (Van + Vbn + Vcn). Unit [V]
12054	Frequency min	Float32	R	Min value of frequency. Unit [Hz]
12056	Ia max	Float32	R	Max of phase A current. Unit [A]
12058	Ib max	Float32	R	Max of phase B current. Unit [A]
12060	Ic max	Float32	R	Max of phase C current. Unit [A]
12062	Iavg max	Float32	R	Max of average current of 3 phase currents. Unit [A]
12064	Residual current max	Float32	R	Max of residual current by sum of 3-phase currents (Ia+ Ib + Ic). Unit [A]
12066	Pa max	Float32	R	Max of active power of phase A. Unit [kW]
12068	Pb max	Float32	R	Max of active power of phase B. Unit [kW]
12070	Pc max	Float32	R	Max of active power of phase C. Unit [kW]
12072	Ptot max	Float32	R	Max of total active power of all phases. Unit [kW]
12074	Qa max	Float32	R	Max of reactive power of phase A. Unit [kVAR]
12076	Qb max	Float32	R	Max of reactive power of phase B. Unit [kVAR]
12078	Qc max	Float32	R	Max of reactive power of phase C. Unit [kVAR]
12080	Qtot max	Float32	R	Max of total reactive power of all phases. Unit [kVAR]
12082	Sa max	Float32	R	Max of apparent power of phase A. Unit [kVA]
12084	Sb max	Float32	R	Max of apparent power of phase B. Unit [kVA]
12086	Sc max	Float32	R	Max of apparent power of phase C. Unit [kVA]
12088	Stot max	Float32	R	Max of total apparent power of all phases. Unit [kVA]
12090-12101	Reserved			
12102	Demand Pa max	Float32	R	Max of active power demand of phase A. Unit [kW]
12104	Demand Pb max	Float32	R	Max of active power demand of phase B. Unit [kW]
12106	Demand Pc max	Float32	R	Max of active power demand of phase C. Unit [kW]
12108	Demand Ptot max	Float32	R	Max of total active power demand of all phases. Unit [kVAR]
12110	Demand Qa max	Float32	R	Max of reactive power demand of phase A. Unit [kVAR]
12112	Demand Qb max	Float32	R	Max of reactive power demand of phase B. Unit [kVAR]
12114	Demand Qc max	Float32	R	Max of reactive power demand of phase C. Unit [kVAR]
12116	Demand Qtot max	Float32	R	Max of total reactive power demand of all phases. Unit [kVAR]
12118	Demand Sa max	Float32	R	Max of apparent power demand of phase A. Unit [kVA]
12120	Demand Sb max	Float32	R	Max of apparent power demand of phase B. Unit [kVA]
12122	Demand Sc max	Float32	R	Max of apparent power demand of phase C. Unit [kVA]
12124	Demand Stot max	Float32	R	Max total apparent power demand of all phases. Unit [kVA]
12126	Demand Ia max	Float32	R	Max of current demand of phase A. Unit [A]
12128	Demand Ib max	Float32	R	Max of current demand of phase B. Unit [A]
12130	Demand Ic max	Float32	R	Max of current demand of phase C. Unit [A]
12132	Demand Iavg max	Float32	R	Max of average current demand of all phases. Unit [A]
12134	THDia max	Float32	R	Max of total harmonic distortion of current Ia. Unit [%]

12136	THDib max	Float32	R	Max of total harmonic distortion of current Ib. Unit [%]
12138	THDic max	Float32	R	Max of total harmonic distortion of current Ic. Unit [%]
12140	TDDia max	Float32	R	Max of total demand distortion of current Ia. Unit [%]
12142	TDDib max	Float32	R	Max of total demand distortion of current Ib. Unit [%]
12144	TDDic max	Float32	R	Max of total demand distortion of current Ic. Unit [%]
12146	Unbal_I max	Float32	R	Max of unbalance ratio of 3 phase currents. Unit [%]
12148	Unbal_U0_I max	Float32	R	Max of current zero-sequence unbalance ratio. Unit [%]
12150	Unbal_U2_I max	Float32	R	Max of current negative-sequence unbalance ratio. Unit [%]
12152	CFa max	Float32	R	Max of crest factor of phase A current.
12154	CFb max	Float32	R	Max of crest factor of phase B current.
12156	CFc max	Float32	R	Max of crest factor of phase C current.
12158	KFa max	Float32	R	Max of K-factor of phase A current.
12160	KFb max	Float32	R	Max of K-factor of phase B current.
12162	KFc max	Float32	R	Max of K-factor of phase C current.
12164	Ia min	Float32	R	Min of phase A current. Unit [A]
12166	Ib min	Float32	R	Min of phase B current. Unit [A]
12168	Ic min	Float32	R	Min of phase C current. Unit [A]
12170	Iavg min	Float32	R	Min of average current of 3 phase currents. Unit [A]
12172	Residual current min	Float32	R	Min of residual current by sum of 3-phase currents (Ia+Ib+Ic). Unit [A]
12174	Pa min	Float32	R	Min of active power of phase A. Unit [kW]
12176	Pb min	Float32	R	Min of active power of phase B. Unit [kW]
12178	Pc min	Float32	R	Min of active power of phase C. Unit [kW]
12180	Ptot min	Float32	R	Min of total active power of all phases. Unit [kW]
12182	Qa min	Float32	R	Min of reactive power of phase A. Unit [kVAR]
12184	Qb min	Float32	R	Min of reactive power of phase B. Unit [kVAR]
12186	Qc min	Float32	R	Min of reactive power of phase C. Unit [kVAR]
12188	Qtot min	Float32	R	Min of total reactive power of all phases. Unit [kVAR]
12190	Sa min	Float32	R	Min of apparent power of phase A. Unit [kVA]
12192	Sb min	Float32	R	Min of apparent power of phase B. Unit [kVA]
12194	Sc min	Float32	R	Min of apparent power of phase C. Unit [kVA]
12196	Stot min	Float32	R	Min of total apparent power of all phases. Unit [kVA]

Measurement Max/Min Data [After user-reset]

Maximum and minimum value among 1 second data measured since user's Max/Min reset (register 10312) point up to now. This Maximum and minimum value are preserved even if power is out.

In case of reading register number 15000 - 15209 by Modbus multi packet, there is no additional data fetch for the packets of increasing register number in order to preserve synchronization of data.

Data fetch is done automatically only for the packets of decreasing register number about register number 15000 - 15209 region.

Register Number	Name	Format	Attribute	Description
15000	Van max	Float32	R	Max of Line to neutral voltage between phase A and neutral. Unit [V]
15002	Vbn max	Float32	R	Max of Line to neutral voltage between phase B and neutral. Unit [V]
15004	Vcn max	Float32	R	Max of Line to neutral voltage between phase C and neutral. Unit [V]
15006	Vavg_In max	Float32	R	Max of Average of 3 line to neutral voltages Van, Vbn, Vcn. Unit [V]
15008	Vab max	Float32	R	Max of Line to line voltage between phase A and B. Unit [V]
15010	Vbc max	Float32	R	Max of Line to line voltage between phase B and C. Unit [V]
15012	Vca max	Float32	R	Max of Line to line voltage between phase C and A. Unit [V]
15014	Vavg_II max	Float32	R	Max of Average of 3 line to line voltages Vab, Vbc, Vca. Unit [V]
15016	THDva max	Float32	R	Max of Total harmonic distortion of line to neutral Van. Unit [%]
15018	THDvb max	Float32	R	Max of Total harmonic distortion of line to neutral Vbn. Unit [%]
15020	THDvc max	Float32	R	Max of Total harmonic distortion of line to neutral Vcn. Unit [%]
15022	Unbal_VIn max	Float32	R	Max of Unbalance ratio of line to neutral voltages. Unit [%]
15024	Unbal_VII max	Float32	R	Max of Unbalance ratio of line to line voltages. Unit [%]
15026	Unbal_U0_V max	Float32	R	Max of voltage zero-sequence unbalance ratio. Unit [%]
15028	Unbal_U2_V max	Float32	R	Max of voltage negative-sequence unbalance ratio. Unit [%]
15030	Temperature max	Float32	R	Max of Ambient temperature of Accura 3300E device. Unit [°C]
15032	Residual voltage max	Float32	R	Max of residual voltage by sum of 3-phase voltages (Van + Vbn + Vcn). Unit [V]
15034	Frequency max	Float32	R	Max value of frequency. Unit [Hz]
15036	Van min	Float32	R	Min of line to neutral voltage between A and neutral. Unit [V]
15038	Vbn min	Float32	R	Min of line to neutral voltage between B and neutral. Unit [V]
15040	Vcn min	Float32	R	Min of line to neutral voltage between C and neutral. Unit [V]
15042	Vavg_In min	Float32	R	Min of average of 3 line to neutral voltages Van, Vbn, Vcn. Unit [V]
15044	Vab min	Float32	R	Min of line to line voltage between A and B. Unit [V]
15046	Vbc min	Float32	R	Min of line to line voltage between B and C. Unit [V]
15048	Vca min	Float32	R	Min of line to line voltage between C and A. Unit [V]
15050	Vavg_II min	Float32	R	Min of average of 3 line to line voltages Vab, Vbc, Vca. Unit [V]

15052	Residual voltage min	Float32	R	Min of residual voltage by sum of 3-phase voltages (Van + Vbn + Vcn). Unit [V]
15054	Frequency min	Float32	R	Min value of frequency. Unit [Hz]
15056	Ia max	Float32	R	Max of phase A current. Unit [A]
15058	Ib max	Float32	R	Max of phase B current. Unit [A]
15060	Ic max	Float32	R	Max of phase C current. Unit [A]
15062	Iavg max	Float32	R	Max of average current of 3 phase currents. Unit [A]
15064	Residual current max	Float32	R	Max of residual current by sum of 3-phase currents (Ia+ Ib + Ic). Unit [A]
15066	Pa max	Float32	R	Max of active power of phase A. Unit [kW]
15068	Pb max	Float32	R	Max of active power of phase B. Unit [kW]
15070	Pc max	Float32	R	Max of active power of phase C. Unit [kW]
15072	Ptot max	Float32	R	Max of total active power of all phases. Unit [kW]
15074	Qa max	Float32	R	Max of reactive power of phase A. Unit [kVAR]
15076	Qb max	Float32	R	Max of reactive power of phase B. Unit [kVAR]
15078	Qc max	Float32	R	Max of reactive power of phase C. Unit [kVAR]
15080	Qtot max	Float32	R	Max of total reactive power of all phases. Unit [kVAR]
15082	Sa max	Float32	R	Max of apparent power of phase A. Unit [kVA]
15084	Sb max	Float32	R	Max of apparent power of phase B. Unit [kVA]
15086	Sc max	Float32	R	Max of apparent power of phase C. Unit [kVA]
15088	Stot max	Float32	R	Max of total apparent power of all phases. Unit [kVA]
15090	PFa max	Float32	R	Max of power factor of phase A.
15092	PFb max	Float32	R	Max of power factor of phase B.
15094	PFc max	Float32	R	Max of power factor of phase C.
15096	Pftot max	Float32	R	Max of total power factor of all phases.
15098	Angle of PFa max	Uint16	R	Angle status of PFa when phase A power factor is maximum. 0: Invalid. Apparent power is equal to zero. 1: Lead angle 2: Lag angle
15099	Angle of PFb max	Uint16	R	Angle status of PFb when phase B power factor is maximum. See "Angle of PFa max". (register 15098)
15100	Angle of PFc max	Uint16	R	Angle status of PFc when phase C power factor is maximum. See "Angle of PFa max". (register 15098)
15101	Angle of Pftot max	Uint16	R	Angle status of Pftot when total power factor is maximum. See "Angle of PFa max". (register 15098)
15102	Demand Pa max	Float32	R	Max of active power demand of phase A. Unit [kW]
15104	Demand Pb max	Float32	R	Max of active power demand of phase B. Unit [kW]
15106	Demand Pc max	Float32	R	Max of active power demand of phase C. Unit [kW]
15108	Demand Ptot max	Float32	R	Max of total active power demand of all phases. Unit [kVAR]
15110	Demand Qa max	Float32	R	Max of reactive power demand of phase A. Unit [kVAR]
15112	Demand Qb max	Float32	R	Max of reactive power demand of phase B. Unit [kVAR]
15114	Demand Qc max	Float32	R	Max of reactive power demand of phase C. Unit [kVAR]

15116	Demand Qtot max	Float32	R	Max of total reactive power demand of all phases. Unit [kVAR]
15118	Demand Sa max	Float32	R	Max of apparent power demand of phase A. Unit [kVA]
15120	Demand Sb max	Float32	R	Max of apparent power demand of phase B. Unit [kVA]
15122	Demand Sc max	Float32	R	Max of apparent power demand of phase C. Unit [kVA]
15124	Demand Stot max	Float32	R	Max total apparent power demand of all phases. Unit [kVA]
15126	Demand Ia max	Float32	R	Max of current demand of phase A. Unit [A]
15128	Demand Ib max	Float32	R	Max of current demand of phase B. Unit [A]
15130	Demand Ic max	Float32	R	Max of current demand of phase C. Unit [A]
15132	Demand Iavg max	Float32	R	Max of average current demand of all phases. Unit [A]
15134	THDia max	Float32	R	Max of total harmonic distortion of current Ia. Unit [%]
15136	THDib max	Float32	R	Max of total harmonic distortion of current Ib. Unit [%]
15138	THDic max	Float32	R	Max of total harmonic distortion of current Ic. Unit [%]
15140	TDDia max	Float32	R	Max of total demand distortion of current Ia. Unit [%]
15142	TDDib max	Float32	R	Max of total demand distortion of current Ib. Unit [%]
15144	TDDic max	Float32	R	Max of total demand distortion of current Ic. Unit [%]
15146	Unbal_I max	Float32	R	Max of unbalance ratio of 3 phase currents. Unit [%]
15148	Unbal_U0_I max	Float32	R	Max of current zero-sequence unbalance ratio. Unit [%]
15150	Unbal_U2_I max	Float32	R	Max of current negative-sequence unbalance ratio. Unit [%]
15152	CFa max	Float32	R	Max of crest factor of phase A current.
15154	CFb max	Float32	R	Max of crest factor of phase B current.
15156	CFc max	Float32	R	Max of crest factor of phase C current.
15158	KFa max	Float32	R	Max of K-factor of phase A current.
15160	KFb max	Float32	R	Max of K-factor of phase B current.
15162	KFc max	Float32	R	Max of K-factor of phase C current.
15164	Ia min	Float32	R	Min of phase A current. Unit [A]
15166	Ib min	Float32	R	Min of phase B current. Unit [A]
15168	Ic min	Float32	R	Min of phase C current. Unit [A]
15170	Iavg min	Float32	R	Min of average current of 3 phase currents. Unit [A]
15172	Residual current min	Float32	R	Min of residual current by sum of 3-phase currents (Ia+Ib+Ic). Unit [A]
15174	Pa min	Float32	R	Min of active power of phase A. Unit [kW]
15176	Pb min	Float32	R	Min of active power of phase B. Unit [kW]
15178	Pc min	Float32	R	Min of active power of phase C. Unit [kW]
15180	Ptot min	Float32	R	Min of total active power of all phases. Unit [kW]
15182	Qa min	Float32	R	Min of reactive power of phase A. Unit [kVAR]
15184	Qb min	Float32	R	Min of reactive power of phase B. Unit [kVAR]
15186	Qc min	Float32	R	Min of reactive power of phase C. Unit [kVAR]
15188	Qtot min	Float32	R	Min of total reactive power of all phases. Unit [kVAR]
15190	Sa min	Float32	R	Min of apparent power of phase A. Unit [kVA]
15192	Sb min	Float32	R	Min of apparent power of phase B. Unit [kVA]
15194	Sc min	Float32	R	Min of apparent power of phase C. Unit [kVA]

15196	Stot min	Float32	R	Min of total apparent power of all phases. Unit [kVA]
15198	PFa min	Float32	R	Min of power factor of phase A.
15200	PFb min	Float32	R	Min of power factor of phase B.
15202	PFc min	Float32	R	Min of power factor of phase C.
15204	Pftot min	Float32	R	Min of total power factor of all phases.
15206	Angle of PFa min	Uint16	R	Angle status of PFa when phase A power factor is minimum. 0: Invalid. Apparent power is equal to zero. 1: Lead angle 2: Lag angle
15207	Angle of PFb min	Uint16	R	Angle status of PFb when phase B power factor is minimum. See "Angle of PFa min". (register 15206)
15208	Angle of PFc min	Uint16	R	Angle status of PFc when phase C power factor is minimum. See "Angle of PFa min". (register 15206)
15209	Angle of Pftot min	Uint16	R	Angle status of Pftot when total power factor is minimum. See "Angle of PFa min". (register 15206)

Harmonics Data

In case of reading register number 13000 - 13384 by Modbus multi packet, there is no additional data fetch for the packets of increasing register number in order to preserve synchronization of data.

Data fetch is done automatically only for the packets of decreasing register number about register number 13000 - 13384 region.

Register Number	Name	Format	Attribute	Description
13000	Harmonics validity	Uint16	R	Register 13001 - 13768 validity. 0: Invalid 1: Valid
13001	DC of voltage phase A	Float32	PR	DC component of phase A voltage. Unit [V]
13003-13064	1st - 31th of voltage phase A	31* Float32	PR	1st - 31th components of phase A voltage. Unit [V]
13065	DC of voltage phase B	Float32	PR	DC component of phase B voltage. Unit [V]
13067-13128	1st - 31th of voltage phase B	31* Float32	PR	1st - 31th components of phase B voltage. Unit [V]
13129	DC of voltage phase C	Float32	PR	DC component of phase C voltage. Unit [V]
13131-13192	1st - 31th of voltage phase C	31* Float32	PR	1st - 31th components of phase C voltage. Unit [V]
13193	DC of current phase A	Float32	PR	DC component of phase A current. Unit [A]
13195-13256	1st - 31th of current phase A	31* Float32	PR	1st - 31th components of phase A current. Unit [A]
13257	DC of current phase B	Float32	PR	DC component of phase B current. Unit [A]
13259-13320	1st - 31th of current phase B	31* Float32	PR	1st - 31th components of phase B current. Unit [A]
13321	DC of current phase C	Float32	PR	DC component of phase C current. Unit [A]
13323-13384	1st - 31th of current phase C	31* Float32	PR	1st - 31th components of phase C current. Unit [A]

Waveform Data

In case of reading register number 14000 - 14793 by Modbus multi packet, there is no additional data fetch for the packets of increasing register number in order to preserve synchronization of data.

Data fetch is done automatically only for the packets of decreasing register number about register number 14000 - 14793 region.

Register Number	Name	Format	Attribute	Description
14000	Validity of 1st cycle Waveform	Uint16	R	Register 14001 - 14396 validity 0: Invalid 1: Valid
14001	Scale factor of voltage phase A	Float32	R	Scale factor with respect to 1st cycle of phase A voltage. The real magnitude of waveform can be obtained by multiplying the scale factor and the waveform.
14003	Scale factor of voltage phase B	Float32	R	Scale factor with respect to 1st cycle of phase B voltage.
14005	Scale factor of voltage phase C	Float32	R	Scale factor with respect to 1st cycle of phase C voltage.
14007	Scale factor of current phase A	Float32	R	Scale factor with respect to 1st cycle of phase A current.
14009	Scale factor of current phase B	Float32	R	Scale factor with respect to 1st cycle of phase B current.
14011	Scale factor of current phase C	Float32	R	Scale factor with respect to 1st cycle of phase C current.
14013-14076	Waveform of voltage phase A	64*Int16	R	1st cycle waveform of phase A voltage. 64-sample / cycle * 1-cycle
14077-14140	Waveform of voltage phase B	64*Int16	R	1st cycle waveform of phase B voltage. 64-sample / cycle * 1-cycle
14141-14204	Waveform of voltage phase C	64*Int16	R	1st cycle waveform of phase C voltage. 64-sample / cycle * 1-cycle
14205-14268	Waveform of current phase A	64*Int16	R	1st cycle waveform of phase A current. 64-sample / cycle * 1-cycle
14269-14332	Waveform of current phase B	64*Int16	R	1st cycle waveform of phase B current. 64-sample / cycle * 1-cycle
14333-14396	Waveform of current phase C	64*Int16	R	1st cycle waveform of phase C current. 64-sample / cycle * 1-cycle
14397	Validity of 2nd cycle Waveform	Uint16	R	Register 14398 - 14793 validity 0: Invalid 1: Valid
14398	Scale factor of voltage phase A	Float32	R	Scale factor with respect to 2nd cycle of phase A voltage. The real magnitude of waveform can be obtained by multiplying the scale factor and the waveform.

14400	Scale factor of voltage phase B	Float32	R	Scale factor with respect to 2nd cycle of phase B voltage.
14402	Scale factor of voltage phase C	Float32	R	Scale factor with respect to 2nd cycle of phase C voltage.
14404	Scale factor of current phase A	Float32	R	Scale factor with respect to 2nd cycle of phase A current.
14406	Scale factor of current phase B	Float32	R	Scale factor with respect to 2nd cycle of phase B current.
14408	Scale factor of current phase C	Float32	R	Scale factor with respect to 2nd cycle of phase C current.
14410-14473	Waveform of voltage phase A	64*Int16	R	2nd cycle waveform of phase A voltage. 64-sample / cycle * 1-cycle
14474-14537	Waveform of voltage phase B	64*Int16	R	2nd cycle waveform of phase B voltage. 64-sample / cycle * 1-cycle
14538-14601	Waveform of voltage phase C	64*Int16	R	2nd cycle waveform of phase C voltage. 64-sample / cycle * 1-cycle
14602-14665	Waveform of current phase A	64*Int16	R	2nd cycle waveform of phase A current. 64-sample / cycle * 1-cycle
14666-14729	Waveform of current phase B	64*Int16	R	2nd cycle waveform of phase B current. 64-sample / cycle * 1-cycle
14730-14793	Waveform of current phase C	64*Int16	R	2nd cycle waveform of phase C current. 64-sample / cycle * 1-cycle

Measurement Event Data Category

Measurement event data category provides fuse fail, real-time event data for voltage phase sequence and over temperature.

Measurement Event Data

Register Number	Name	Format	Attribute	Description
10380	Fuse fail of phase A	Uint16	R	Status of A phase voltage is 0, A phase current is not 0 at the same time. 0: Normal 1: Fuse fail
10381	Fuse fail of phase B	Uint16	R	Status of B phase voltage is 0, B phase current is not 0 at the same time. 0: Normal 1: Fuse fail
10382	Fuse fail of phase C	Uint16	R	Status of C phase voltage is 0, C phase current is not 0 at the same time. 0: Normal 1: Fuse fail
10383	Phase open of phase A	Uint16	R	Status of A phase voltage and current are 0. 0: Normal 1: Phase open
10384	Phase open of phase B	Uint16	R	Status of B phase voltage and current are 0. 0: Normal 1: Phase open
10385	Phase open of phase C	Uint16	R	Status of C phase voltage and current are 0. 0: Normal 1: Phase open
10386	Over temperature	Uint16	R	Over temperature measured from side of Accura 3300E. 0: Normal 1: Over temperature
10387	Status of Event LED	Uint16	R	Status of LED light by event occur. 0: Off 1: On
10388	Status of LCD backlight	Uint16	R	Status of LCD backlight by Event occur. 0: Off 1: On

Chapter 3 Modbus Map Application

Register Addressing

Holding registers are addressed starting at zero: Register address is obtained by subtracting by 1 from register number of Modbus Map. Holding registers 1 - 65536 are addressed as 0 - 65535

For example, the request packet for reading "Voltage Vab" (register number 11030 - 11031) is follows. (11030-1 → 2B15h).

Request packet		
03h	2B15h	0002h
Function Code (1 byte)	Starting Address (2 bytes)	Quantity of Registers (2 bytes)

Data Format

Measurement data types used in Accura 3300E are as follows.

Data format	Description	Word Length	Endian	Range
UInt16	Unsigned 16-bit	1	NA ¹	0 to 65,535
Int16	Signed 16-bit	1	NA ¹	-32,768 to 32,767
UInt32	Unsigned 32-bit	2	Big-Endian ²	0 to 4,294,967,295
Int32	Signed 32-bit	2	Big-Endian ²	-2,147,483,648 to 2,147,483,647
Float32	Single-precision Float (IEEE 754)	2	Big-Endian ²	-3.4x10 ³⁸ to 3.4x10 ³⁸

1. NA (Not Available): 1-word data, independent of endian.

2. 2-word data, two register spaces are used. The upper word is located in the lower address register and the lower word is located in the higher address register.

Endian

A measured parameter that has 2 word-length attribute such as "UInt32", "Int32" or "Float32" needs 2 register spaces in Modbus map. Accura 3300E supports "Big-Endian". The high-order word is in the lower register and the low-order word is in the higher addressed register.

For example, suppose that the measured line to line "Voltage Vab" that has float32 attribute (register number 11030 - 11031) is 380.2 Volts.

(Decimal) 380.2 → (Float32) 43BE1999h

Register number	Name	Value	Remarks
11030	Voltage Vab	43BEh	High-order word of Vab
11031		1999h	Low-order word of Vab

Setup of device

Setup in remote setup mode is locked by default. At first, it is necessary to unlock the setup at the Modbus connection

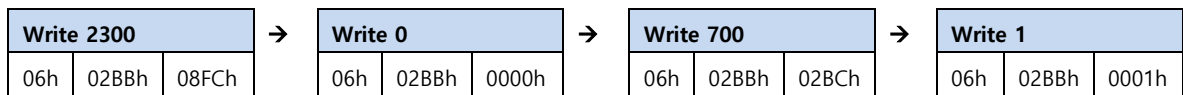
Remote Setup Unlocking

For setup to be allowed, write these four numbers on the register 10100 in the order given below.

Write 2300 → Write 0 → Write 700 → Write 1

If the order is wrong, the entire writing process should start from the beginning again.

699(700-1)→02BBh, 700→02BCh, 2300→08FCh



Remote Setup Locking

For setup to be locked, write any value on the register 10100.

Write 0		
06h	02BBh	0000h

Setup-lock status can be known by reading this register as shown below:

- 1: (default) Setup locked
- 0: Setup Allowed

Control of device

Remote control is locked by default. At first, it is necessary to unlock the control at the Modbus connection in which the change is to be made.

Remote Control Unlocking

For control to be allowed, write these four numbers on the register 10300 in the order given below.

Write 2300 → Write 0 → Write 1600 → Write 1

If the order is wrong, the entire writing process should start from the beginning again.

1599(1600-1)→063Fh, 1600→0640h, 2300→08FCh



Remote Control Locking

For control to be locked, write any value on the register 10300.

Write 0		
06h	063Fh	0000h

Control-lock status can be known by reading this register as shown below:

- 1: (default) Control locked
- 0: Control Allowed

APPENDIX A Sample of Modbus RTU Packet

The following is a Modbus RTU packet sample in which Modbus registers 1 - 3 are read using Function code 03h "Read holding registers". Modbus registers 1 - 3 are addressed 0 - 2 by subtracting by 1. It is assumed that the "Device Address" of Accura 3300E is 1.

Request Packet

Device Address	Function Code	Data		CRC
		Starting Address	Quantity of Registers	
1 byte	1 byte	2 bytes	2 bytes	2 bytes
01h	03h	0000h	0003h	05CBh

CRC: See Appendix B for how to get the CRC. (CRC high-order byte is the last byte to be sent)

Response Packet

Device Address	Function Code	Data				CRC
		Byte Count	Quantity of Registers			
1 byte	1 byte	1 byte	6 bytes			2 bytes
01h	03h	06h	0CE5h	ABCDh	1234h	C433h

CRC: See Appendix B for how to get the CRC. (CRC high-order byte is the last byte to be sent)

APPENDIX B CRC-16[Modbus] Algorithm

CRC Table

```
unsigned int CrcTable[256];
unsigned int GenCrc(unsigned int Data, unsigned int Polynomial, unsigned int crc) {
    unsigned int i;
    for(i = 0; i < 8; i++) {
        if((Data ^ crc) & 1){
            crc = (crc >> 1) ^ Polynomial;
        } else {
            crc >>= 1;
        }
        Data >>= 1;
    }
    return (crc & 0xFFFF);
}

void MakeCrcTable() {
    unsigned int Polynomial = 0xA001;
    unsigned int i;
    for(i = 0; i < 256; i++)
        CrcTable[i] = GenCrc(i, Polynomial, 0);
}
```

CRC Function

```
unsigned int CRC16(unsigned char *puchMsg, unsigned short usDataLen) {
    unsigned char uchCRCHi = 0xFF;
    unsigned char uchCRCLo = 0xFF;
    unsigned ulIndex;
    while(usDataLen--) {
        ulIndex = uchCRCHi ^ *puchMsg++;
        uchCRCHi = uchCRCLo ^ (CrcTable[ulIndex] & 0xFF);
        uchCRCLo = (CrcTable[ulIndex] >> 8) & 0xFF;
    }
    return ((uchCRCHi << 8) | uchCRCLo);
}
```

APPENDIX C Accura 3300 Modbus Map Support

Accura 3300E provides convenience of accessibility to Accura 3300 users by having compatibility field on Modbus Map. However, does not provide compatibility for some field.

Modbus Map Overview

Register Number	Section	Description
1-15	System Information	Product model, basic module, extension module, version, calibration
51-66	Setup	Communication id, PT/CT ratio, protocol, baud rate, parity bit, stop bit, demand time, extension module selection
101-147	Measurement	Voltage, current, fundamental current, kW, kVAR, kVA, kWh, kVARh, kVAh, PF, frequency
148-156	THD, K-factor	Voltage THD, current THD, current K-factor
157-168	Energy	kWh/kVARh received, kWh/kVARh delivered, kWh/kVARh total
301-367	Demand, Maximum, Minimum	Demand, peak demand, maximum, minimum
401-592	Harmonics	Voltage harmonics, current harmonics
593-605	Vector Diagram	Voltage[x, y], current[x, y]
608-997	Waveform	Voltage waveform, current waveform
1001-1097	Demand Trend	Total kW demand trend
1101-1107	Reset	kWh reset, kVARh reset, kVAh reset, all demand reset, all peak demand reset, max/min reset, offset reset
9001-9074	Short-formed data	Collection of measurements and controls

System Information Section

Register Number	Name	Format	Attribute	Description
1	Product model	Uint16	R	Accura 3300E Product ID: 3301
2	Serial number	Uint32	R	Serial number
10	Hardware version	Uint16	R	Hardware version
11	Firmware version	Uint16	R	Firmware version
12	Map version	Uint16	R	Map version
13	Calibration year	Uint16	R	Calibration year
14	Calibration month	Uint16	R	Calibration month
15	Calibration date	Uint16	R	Calibration date

Setup Section

Register Number	Name	Format	Attribute	Description
51	Communication ID	Uint16	RW	Modbus serial address. (Slave) Valid range: 1 - 247 Default: 1
52	Wiring mode	Uint16	RW	0: 1P2W, 1-phase 2-wire voltage connection 1: 1P3W, 1-phase 3-wire voltage connection 2: 3P3W, 3-phase 3-wire voltage connection 3: (default) 3P4W, 3-phase 4-wire voltage connection
53	PT ratio	Uint16	RW	PT ratio = PT Primary/ PT Secondary Real PT ratio = (PT ratio) * 0.1 Valid range: 0 - 9999 Default: 10
54	CT ratio	Uint16	RW	CT ratio = CT Primary/ CT Secondary Valid range: 0 - 5000 Default: 10
55	Protocol	Uint16	RW	0: Rootech protocol 1: (default) Modbus RTU
56	Baud rate	Uint16	RW	0: 1200 1: 2400 2: 4800 3: (default) 9600 4: 19200 5: 38400 6: 57600 7: 115200
57	Parity bit	Uint16	RW	0: None parity 1: Odd parity 2: Even parity
58	Stop bit	Uint16	RW	0: 1-stop bit 1: 2-stop bit
62	Calculation method ¹ for reactive power	Uint16	RW	0: (default) Fundamental calculation 1: Harmonic calculation
63	Demand time	Uint16	RW	Valid range: 1 - 60 [min] Default: 15 [min]
65	Common use frequency	Uint16	R	0: 60Hz 1: 50Hz
66	Selection of energy display type	Uint16	RW	0: kWh/kVArh Received. Energy flows from source to load 1: kWh/kVArh Delivered. Energy flows from load to source 2: kWh/kVArh Total 3: kWh/kVArh Net: (kWh Received) - (kWh Delivered)

1. Refer to "Accura 3300E User Manual" for detail information of method.

Measurement Section

Measurement field 101 - 147 is the region for providing the measurement data of the device. In order to obtain final data, should multiply the measured value and scale by using scale register.

Register Number	Name	Format	Attribute	Description
101	Voltage Van	Uint16	R	This value should be multiplied by (voltage scale * 0.1). For voltage scale, see register 109. Unit [V]
102	Voltage Vbn	Uint16	R	Same as above.
103	Voltage Vcn	Uint16	R	Same as above.
104	Phase Voltage Average Vavg_In	Uint16	R	Same as above.
105	Line Voltage Vab	Uint16	R	Same as above.
106	Line Voltage Vbc	Uint16	R	Same as above.
107	Line Voltage Vca	Uint16	R	Same as above.
108	Line Voltage Average Vavg_LL	Uint16	R	Same as above.
109	Voltage scale	Uint16	R	Used in registers 101 - 108.
110	Current Ia	Uint16	R	This value should be multiplied by (current scale * 0.001). For current scale, see register 118. Unit [A]
111	Current Ib	Uint16	R	Same as above.
112	Current Ic	Uint16	R	Same as above.
113	Current Average Iavg	Uint16	R	Same as above.
114	Fundamental Current Ia1	Uint16	R	This value should be multiplied by (current scale * 0.001). For current scale, see register 118. Unit [A]
115	Fundamental Current Ib1	Uint16	R	Same as above.
116	Fundamental Current Ic1	Uint16	R	Same as above.
117	Fundamental Current Average Iavg1	Uint16	R	Same as above.
118	Current scale	Uint16	R	Used in registers 110 - 117.
119	kW Pa	Int16	R	This value should be multiplied by (kW scale * 0.001). For kW scale, see register 122. Unit [kW]
120	kW Pb	Int16	R	Same as above.
121	kW Pc	Int16	R	Same as above.
122	kW scale	Uint16	R	Used in registers 119 - 121.
123	Total kW Ptot	Int16	R	This value should be multiplied by (Total kW scale * 0.001). For total kW scale, see register 124. Unit [kW]
124	Total kW scale	Uint16	R	Used in register 123.
125	kVAR Qa	Int16	R	This value should be multiplied by (kVAR scale * 0.001). For kVAR scale, see register 128. Unit [kVAR]

126	kVAR Qb	Int16	R	This value should be multiplied by (kVAR scale * 0.001). For kVAR scale, see register 128. Unit [kVAR]
127	kVAR Qc	Int16	R	Same as above.
128	kVAR scale	Uint16	R	Used in registers 125 - 127.
129	Total kVAR Qtot	Int16	R	This value should be multiplied by (Total kVAR scale * 0.001). For total kVAR scale, see register 130. Unit [kVAR]
130	Total kVAR scale	Uint16	R	Used in register 129.
131	kVA Sa	Int16	R	This value should be multiplied by (kVA scale * 0.001). For kVA scale, see register 134. Unit [kVA]
132	kVA Sb	Int16	R	Same as above.
133	kVA Sc	Int16	R	Same as above.
134	kVA scale	Uint16	R	Used in registers 131 - 133.
135	Total kVA Stot	Int16	R	This value should be multiplied by (Total kVA scale * 0.001). For total kVA scale, see register 136. Unit [kVA]
136	Total kVA scale	Uint16	R	Used in register 135.
137	PF A	Int16	R	This value should be multiplied by 0.001.
138	PF B	Int16	R	This value should be multiplied by 0.001.
139	PF C	Int16	R	This value should be multiplied by 0.001.
140	Total Pftot	Int16	R	This value should be multiplied by 0.001.
141	Frequency.	Uint16	R	This value should be multiplied by 0.01. Unit [Hz]
142	kWh net	Int32	R	kWh net: (kWh received - kWh delivered). Unit [kWh]
144	kVARh net	Int32	R	kVARh net: (kVARh received - kVARh delivered). Unit [kVARh]
146-147	kVAh.	Int32	R	Unit [kVAh]

THD, K-factor Section

Register Number	Name	Format	Attribute	Description
148	THD of voltage Va	Uint16	R	This value should be multiplied by 0.1. Unit [%]
149	THD of voltage Vb	Uint16	R	Same as above.
150	THD of voltage Vc	Uint16	R	Same as above.
151	THD of current Ia	Uint16	R	This value should be multiplied by 0.1. Unit [%]
152	THD of current Ib	Uint16	R	Same as above.
153	THD of current Ic	Uint16	R	Same as above.
154	K-factor of current Ia	Uint16	R	This value should be multiplied by 0.01. Unit [%]
155	K-factor of current Ib	Uint16	R	Same as above.
156	K-factor of current Ic	Uint16	R	Same as above.

Energy Section

Register Number	Name	Format	Attribute	Description
157	kWh received	Uint32	RW	
159	kWh delivered	Uint32	RW	
161	kWh total	Uint32	R	kWh total: kWh received + kWh delivered.
163	kVARh received.	Uint32	RW	
165	kVARh delivered.	Uint32	RW	
167-168	kVARh total	Uint32	R	kVARh total: kVARh received + kVARh delivered.

Demand, Maximum, Minimum Section

Register Number	Name	Format	Attribute	Description
301	Total kW demand	Int16	R	This value should be multiplied by (Total kW demand scale * 0.001). For total kW demand scale, see register 302. Unit [kW]
302	Total kW demand scale	Uint16	R	Used in register 301.
303	Total kVAR demand	Int16	R	This value should be multiplied by (Total kVAR demand scale * 0.001). For total kVAR demand scale, see register 304. Unit [kVAR]
304	Total kVAR demand scale	Uint16	R	Used in register 303.
305	Total kVA demand	Int16	R	This value should be multiplied by (Total kVA demand scale * 0.001). For total kVA demand scale, see register 306. Unit [kVA]
306	Total kVA demand scale	Uint16	R	Used in register 305.
307	Current demand of Ia	Uint16	R	This value should be multiplied by (Current demand scale * 0.001). For current demand scale, see register 311. Unit [A]
308	Current demand of Ib	Uint16	R	Same as above.
309	Current demand of Ic	Uint16	R	Same as above.
311	Current demand scale	Uint16	R	Used in registers 307 - 309.
312	Total kW peak demand	Int16	R	This value should be multiplied by (Total kW peak demand scale * 0.001). For total peak kW demand scale, see register 313. Unit [kW]
313	Total kW peak demand scale	Uint16	R	Used in register 312.
314	Total kVAR peak demand	Int16	R	This value should be multiplied by (Total kVAR peak demand scale * 0.001). For total kVAR peak demand scale, see register 315. Unit [kVAR]
315	Total kVAR peak demand scale	Uint16	R	Used in register 314.
316	Total kVA peak demand	Int16	R	This value should be multiplied by (Total kVA peak demand scale * 0.001). For total kVA peak demand scale, see register 317. Unit [kVA]

317	Total kVA peak demand scale	Uint16	R	Used in register 316.
318	Current peak demand of Ia	Uint16	R	This value should be multiplied by (current peak demand scale * 0.001). For current peak demand scale, see register 322. Unit [A]
319	Current peak demand of Ib	Uint16	R	Same as above.
320	Current peak demand of Ic	Uint16	R	Same as above.
322	Current peak demand scale	Uint16	R	Used in registers 318 - 320.
323	Voltage Van maximum	Uint16	R	This value should be multiplied by (voltage maximum scale * 0.1). For voltage maximum scale, see register 331. Unit [V]
324	Voltage Vbn maximum	Uint16	R	Same as above.
325	Voltage Vcn maximum	Uint16	R	Same as above.
326	Voltage average Vavg_In maximum	Uint16	R	Same as above.
327	Line voltage Vab maximum	Uint16	R	This value should be multiplied by (voltage maximum scale * 0.1). For voltage maximum scale, see register 331. Unit [V]
328	Line voltage Vbc maximum	Uint16	R	Same as above.
329	Line voltage Vca maximum	Uint16	R	Same as above.
330	Line voltage average Vavg_II maximum	Uint16	R	Same as above.
331	Voltage maximum scale	Uint16	R	Used in registers 323 - 330.
332	Current Ia maximum	Uint16	R	This value should be multiplied by (current maximum scale * 0.001). For current maximum scale, see register 340. Unit [A]
333	Current Ib maximum	Uint16	R	Same as above.
334	Current Ic maximum	Uint16	R	Same as above.
335	Current Iavg maximum	Uint16	R	Same as above.
336	Fundamental Current Ia maximum	Uint16	R	This value should be multiplied by (current maximum scale * 0.001). For current maximum scale, see register 340. Unit [A]
337	Fundamental Current Ib maximum	Uint16	R	Same as above.
338	Fundamental Current Ic maximum	Uint16	R	Same as above.
339	Fundamental Current Iavg maximum	Uint16	R	Same as above.
340	Current maximum scale	Uint16	R	Used in registers 332 - 339.
341	kW Pa maximum	Int16	R	This value should be multiplied by (kW maximum scale * 0.001). For kW maximum scale, see register 344. Unit [kW]
342	kW Pb maximum	Int16	R	Same as above.
343	kW Pc maximum	Int16	R	Same as above.

344	kW maximum scale	Uint16	R	Used in registers 341 - 343.
345	Total kW maximum	Int16	R	This value should be multiplied by (Total kW maximum scale * 0.001). For total kW maximum scale, see register 346. Unit [kW]
346	Total kW maximum scale	Uint16	R	Used in register 345.
347	kVAR Qa maximum	Int16	R	This value should be multiplied by (kVAR maximum scale * 0.001). For kVAR maximum scale, see register 351. Unit [kVAR]
348	kVAR Qb maximum	Int16	R	Same as above.
349	kVAR Qc maximum	Int16	R	Same as above.
350	kVAR maximum scale	Uint16	R	Used in registers 347 - 349.
351	Total kVAR maximum	Int16	R	This value should be multiplied by (Total kVAR maximum scale * 0.001). For total kVAR maximum scale, see register 352. Unit [kVAR]
352	Total kVAR maximum scale	Uint16	R	Used in register 351.
353	kVA Sa maximum	Int16	R	This value should be multiplied by (kVA maximum scale * 0.001). For kVA maximum scale, see register 356. Unit [kVA]
354	kVA Sb maximum	Int16	R	Same as above.
355	kVA Sc maximum	Int16	R	Same as above.
356	kVA maximum scale	Uint16	R	Used in registers 353 - 355.
357	Total kVA maximum	Int16	R	This value should be multiplied by (Total kVA maximum scale * 0.001). For total kVA maximum scale, see register 358. Unit [kVA]
358	Total kVA maximum scale	Uint16	R	Used in register 357.
359	Voltage Van minimum	Uint16	R	This value should be multiplied by (voltage minimum scale * 0.1). For voltage minimum scale, see register 367. Unit [V]
360	Voltage Vbn minimum	Uint16	R	Same as above.
361	Voltage Vcn minimum	Uint16	R	Same as above.
362	Voltage Vavg_In minimum	Uint16	R	Same as above.
363	Line Voltage Vab minimum	Uint16	R	This value should be multiplied by (voltage minimum scale * 0.1). For voltage minimum scale, see register 367. Unit [V]
364	Line Voltage Vbc minimum	Uint16	R	Same as above.
365	Line Voltage Vca minimum	Uint16	R	Same as above.
366	Line Voltage average Vavg_II minimum	Uint16	R	Same as above.
367	Voltage minimum scale	Uint16	R	Used in registers 359 - 366.

Harmonic Section

The ratio of each harmonic is the percentage with respect to fundamental wave.

Register Number	Name	Format	Attribute	Description
401	DC ratio of voltage Va	Uint16	R	This value should be multiplied by 0.1. Unit [%]
402-432	1st - 31th harmonic ratio of voltage Va	31* Uint16	R	Same as above.
433	DC ratio of voltage Vb	Uint16	R	This value should be multiplied by 0.1. Unit [%]
434-464	1st - 31th harmonic ratio of voltage Vb	31* Uint16	R	Same as above.
465	DC ratio of voltage Vc	Uint16	R	This value should be multiplied by 0.1. Unit [%]
466-496	1st - 31th harmonic ratio of voltage Vc	31* Uint16	R	Same as above.
497	DC ratio of current Ia	Uint16	R	This value should be multiplied by 0.1. Unit [%]
498-528	1st - 31th harmonic ratio of current Ia	31* Uint16	R	Same as above.
529	DC ratio of current Ib	Uint16	R	This value should be multiplied by 0.1. Unit [%]
530-560	1st - 31th harmonic ratio of current Ib	31* Uint16	R	Same as above.
561	DC ratio of current Ic	Uint16	R	This value should be multiplied by 0.1. Unit [%]
562-592	1st - 31th harmonic ratio of current Ic	31* Uint16	R	Same as above.

Vector Diagram Section

Present voltage and current as relative x, y coordinates after put the case of voltage of phase A is zero.

Register Number	Name	Format	Attribute	Description
593	Voltage Vax	Int16	R	This value should be multiplied by 0.1. Unit [V]
594	Voltage Vay	Int16	R	Same as above.
595	Voltage Vbx	Int16	R	Same as above.
596	Voltage Vby	Int16	R	Same as above.
597	Voltage Vcx	Int16	R	Same as above.
598	Voltage Vcy	Int16	R	Same as above.
600	Current Iax	Int16	R	This value should be multiplied by 0.001. Unit [A]
601	Current Iay	Int16	R	Same as above.
602	Current Ibx	Int16	R	Same as above.
603	Current Iby	Int16	R	Same as above.
604	Current Icx	Int16	R	Same as above.
605	Current Icy	Int16	R	Same as above.

Waveform Section

3 Phase voltage and current are sampled by 12-bit ADC which is inner part of Accura 3300E. 12-bit ADC afford 64 sampling data as 1-cycle waveform.

Register Number	Name	Format	Attribute	Description
608	Waveform Update Flag	Uint16	R	Write[0xFF]: Update Request Read[0x00]: Update Done
609-672	1st - 64th ADC raw sample of voltage Va	64* Int16	R	Waveform of phase A voltage.
674-737	1st - 64th ADC raw sample of current Ia	64* Int16	R	Waveform of phase A current.
739-802	1st - 64th ADC raw sample of voltage Vb	64* Int16	R	Waveform of phase B voltage.
804-867	1st - 64th ADC raw sample of current Ib	64* Int16	R	Waveform of phase B current.
869-932	1st - 64th ADC raw sample of voltage Vc	64* Int16	R	Waveform of phase C voltage.
934-997	1st - 64th ADC raw sample of current Ic	64* Int16	R	Waveform of phase C current.

Demand trend Section

Register Number	Name	Format	Attribute	Description
1001	Total kW demand trend value1	Int16	R	This value should be multiplied by (total kW demand trend value scale * 0.001). For total kW demand trend value scale, see register 1097. Unit [kW]
1002-1096	Total kW demand trend value 2 - 96	95* Int16	R	Each value should be multiplied by (total kW demand trend value scale * 0.001). For total kW demand trend value scale, see register 1097. Unit [kW]
1097	Total kW demand trend value scale	Uint16	R	Used in registers 1001-1096.

Reset Section

Register Number	Name	Format	Attribute	Description
1101	kWh Reset	Uint16	W	Write 0x00FF: kWh Reset
1102	kVARh Reset	Uint16	W	Write 0x00FF: kVARh Reset
1103	kVAh Reset	Uint16	W	Write 0x00FF: kVAh Reset
1104	All Demand Reset	Uint16	W	Write 0x00FF: All Demand Reset
1105	All Peak Demand Reset	Uint16	W	Write 0x00FF: All Peak Demand Reset
1106	Max/Min Reset	Uint16	W	Write 0x00FF: Max/Min Reset

Short-formed Data Section

Short-formed data section supports floating point format for most data.

Register Number	Name	Format	Attribute	Description
9001	Voltage Van	Float32	R	Line to neutral voltage between phase A and neutral. Unit [V]
9003	Voltage Vbn	Float32	R	Line to neutral voltage between phase B and neutral. Unit [V]
9005	Voltage Vcn	Float32	R	Line to neutral voltage between phase C and neutral. Unit [V]
9007	Voltage Vab	Float32	R	Line to line voltage between phase A and B. Unit [V]
9009	Voltage Vbc	Float32	R	Line to line voltage between phase B and C. Unit [V]
9011	Voltage Vca	Float32	R	Line to line voltage between phase C and A. Unit [V]
9013	Current Ia	Float32	R	Current of phase A. Unit [A]
9015	Current Ib	Float32	R	Current of phase B. Unit [A]
9017	Current Ic	Float32	R	Current of phase C. Unit [A]
9021	kW Pa	Float32	R	Active power of phase A. Unit [kW]
9023	kW Pb	Float32	R	Active power of phase B. Unit [kW]
9025	kW Pc	Float32	R	Active power of phase C. Unit [kW]
9027	Total kW	Float32	R	Total active power of all phases. Unit [kW]

9029	kVAR Qa	Float32	R	Reactive power of phase A. Unit [kVAR]
9031	kVAR Qb	Float32	R	Reactive power of phase B. Unit [kVAR]
9033	kVAR Qc	Float32	R	Reactive power of phase C. Unit [kVAR]
9035	Total kVAR	Float32	R	Total reactive power of all phases. Unit [kVAR]
9037	kVA Sa	Float32	R	Apparent power of phase A. Unit [kVA]
9039	kVA Sb	Float32	R	Apparent power of phase B. Unit [kVA]
9041	kVA Sc	Float32	R	Apparent power of phase C. Unit [kVA]
9043	Total kVA	Float32	R	Total apparent power of all phases. Unit [kVA]
9045	PF a	Int16	R	Power factor of phase A. This value should be multiplied by 0.001.
9046	PF b	Int16	R	Power factor of phase B. This value should be multiplied by 0.001.
9047	PF c	Int16	R	Power factor of phase C. This value should be multiplied by 0.001.
9048	Total PF	Int16	R	Total power factor of all phases. This value should be multiplied by 0.001.
9049	Frequency	Uint16	R	Frequency of input voltage. This value should be multiplied by 0.01. Unit [Hz]
9050	kWh	Int32	R	kWh energy of total active power. Unit [kWh]
9052	kVARh	Int32	R	kVAh energy of total apparent power. Unit [kVAh]
9054	THD of voltage Va	Uint16	R	Total harmonic distortion of line to neutral Van. This value should be multiplied by 0.1. Unit [%]
9055	THD of voltage Vb	Uint16	R	Total harmonic distortion of line to neutral Vbn. This value should be multiplied by 0.1. Unit [%]
9056	THD of voltage Vc	Uint16	R	Total harmonic distortion of line to neutral Vcn. This value should be multiplied by 0.1. Unit [%]
9057	THD of current Ia	Uint16	R	Total harmonic distortion of current Ia. This value should be multiplied by 0.1. Unit [%]
9058	THD of current Ib	Uint16	R	Total harmonic distortion of current Ib. This value should be multiplied by 0.1. Unit [%]
9059	THD of current Ic	Uint16	R	Total harmonic distortion of current Ic. This value should be multiplied by 0.1. Unit [%]
9074	kWh/kVARh Reset	Uint16	W	Write 0x0001: Reset

Accura 3300E

Communication User Guide

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